

FFAR Final Progress Report

As part of closing out a FFAR grant, grantees must complete a Final Progress Report. Please use the template below to complete the programmatic report. The report must be submitted to FFAR within 90 days after the expiration or termination of a FFAR funded research grant. All questions about this form should be directed to grants@foundationfar.org.

The Final Progress Report communicates the cumulative results and major accomplishments of the funded grant research, including accomplishment for each aim and whether all research aims were fully completed. FFAR uses the content of the Final Progress Report to inform Congress, the Advisory Board, and other stakeholders on the successes, impacts and value of funding unique partnerships to support innovative science addressing today's food and agriculture challenges. If a question is not applicable to the project, please type "Nothing to report."

Grant Information	
Grant ID	593608
Award Program	New Innovator in Food and Agriculture
Project Title	Harnessing Endophytes to Improve Crop Efficiency and Production
Grant Start Date	01/01/2019
Grant End Date	12/31/2023
Total Grant Budget	\$584,460.82
Total FFAR Budget	\$292,230.41
Matching Funder(s)	University of Georgia
Final Report Date	12/31/2023
Project Director/Principal Investigator Information	
Full Name	Jason Wallace
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Grantee Organization Information	
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Tax ID	58-1353149



Authorized Signing Official (ASO) Information	
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General Information

- 1.1. Please list the geographic location(s) – city, state, congressional district - where the work was conducted. If the work was conducted outside of the US, please list the city and country.

Athens, Georgia, USA (Georgia 10th congressional district)

- 1.2. How many new jobs were created because of this grant funding?

2

- 1.3. How many jobs were maintained because of this grant funding?

4

- 1.4. Have there been any changes to your organization's IRS 501(c)(3) non-profit status since you were awarded the grant? If yes, please explain.

No

- 1.5. Has your organization undergone a recent name change? If so, please provide the new name of your organization.

No

2. Scientific Report

- 2.1. Public Abstract (up to 500 words): The abstract should be 'stand-alone' and is intended for a general audience. It should be a concise overview/summary of the importance of the project, the issues that the project addressed, and the key findings of the project. The abstract should also clearly state how the results of the project help to address important needs in U.S. food and agriculture systems.

"Endophytes" are microbes that live inside plant tissue. Although some can cause disease,

many are harmless and some even benefit the plant. This project aimed to understand how beneficial endophytes work with corn (maize, *Zea mays*) to lay the groundwork for better harnessing these interactions for improved agriculture. Our results indicate that these beneficial interactions rely on having the right combination of genetics from both the host and the microbe side. (Meaning, a beneficial microbe generally affects different maize varieties in different ways. Conversely, a single maize variety will respond differently depending on which microbe it's treated with.) When looking across the United States, we found that the variety of maize had little consistent effect on which microbes live in its stalks, and that the environment was much more significant. However, plant genetics did have a significant effect within a given environment. This "gene-by-environment interaction" means that plant-microbe relationships can be very complex, depending on the combination of plant variety, environmental conditions, and microbes present in that environment. Finally, we find a small number of differences between inbred and hybrid maize, indicating little role for heterosis (hybrid vigor) for the maize microbiome. The environment and especially the tissue seemed to have significantly stronger effects. Overall, our results indicate that the microenvironment (plant tissue) seems to be the strongest driver of the microbial community, followed by the macroenvironment (geographic location and parameters), with plant genetics playing a significant role within a given environment but showing little to no consistency across environments.

- 2.2. Provide a description or interpretation of how the key findings of the project can be used by the agriculture community. It is important the PI identifies how these findings have been, or may be, adopted or adapted in U.S. agriculture and food systems. If the findings cannot yet be applied, this section should address how they can be used to guide future planning or decision-making (up to 1,000-word limit).

These findings have been broken down by the major aims of the project:

Objective 1 – Genetic variability of maize's response to beneficial endophytes

Our first Aim investigated how genetic variation among maize varieties affects how they interact with beneficial endophytes. For the agriculture industry, the first take-home is that the maize variety they plant has a much larger impact on traits (like yield) than whether they apply endophytes. This is what we expected going in; however, in looking at variability across varieties we found little to no consistency in how endophytes affected the different varieties. Some boosted one variety while hurting another, some affected shoot growth in one variety but root growth in another, and in almost all cases the effects were small and prone to noise. The ultimate take-home lesson from this is that even though endophytes can impact plant growth, they are not a one-size-fits-all solution, and they may or may not work on a particular maize variety. While we did not explicitly test commercial treatments as part of this study, our conversations with farmers and other researchers indicates that commercial treatments are similarly variable and unreliable (though some of that is probably also due to environmental variability, below.)

Objective 2 – Gene x Environment interaction regarding maize endophytes across the US

Our second aim looked at the natural microbiome in maize stalks across the US. This was a collaboration with the Genomes to Fields Initiative, a large, grassroots collaborations of 50+ researchers across the country. By sampling the same varieties of maize at 20 locations, we were able to parse apart how large of an effect genetics and the environment had on these communities. We found that environment was a large driver of these communities, with genetics having very little consistent effect by itself. The combination of genetics within an environment, however, was at least as strong as environment by itself. In practice, this means that a given variety will have very different microbes when you put it in different locations, but once you know what it's like at one location, it tends to be more consistent. For producers, this ties in to the genetic variability above (Aim 1), in that the local environment seems to have a large impact on both the microbes themselves and on how they interact with plants. This is another potential source of the variability (=unreliability) several growers have mentioned to us regarding commercial microbial treatments. Though we did not specifically test commercial seed treatments, the implication is that their effectiveness will likely change based on both where they are used and what the specific environment (heat, rainfall, management) is like that year.

Objective 3 – The effect of inbreeding on the maize microbiome

Maize shows heterosis (hybrid vigor) for essentially every agronomic trait, and this aim meant to see what heterosis looks like when applied to the microbiome. Our goal was to compare inbred and hybrid maize in three different environments to see if there were any consistent differences that would indicate heterotic effects. We found only very few microbes that were reliably different between inbreds and hybrids, implying little to no heterosis for the general microbial community. We did see that plant tissue and environment were much stronger drivers. Interestingly, the stalk microbiome converged across the different environments despite the soil and root microbiomes being very different. This implies that the stalk is a strong filter for the microbes that can live there. This project was geared toward very basic science and so does not have any immediate impact for farmers, but will likely have implications for those developing beneficial microbes. The most important take-away is that few organisms are able to make it into the maize stalk, and the community is relatively consistent across environments. This implies that (1) a beneficial microbe able to live in the stalk may do so consistently across environments, and (2) beneficial microbes that stay confined to the root or rhizosphere may be easier to develop since these compartments seem more amenable to a large number of diverse microbes.

Other lessons learned

- 2.3. Research Outcomes/Impact (up to 1,500-word limit). The primary goal here is to answer questions such as "How did this project lead to improvements in U.S. agriculture and food systems?" or "How can the findings of this study guide or make improvement in future investigations and research?" Outcomes should be explained and classified in one of the following ways:
- a) potential outcomes, i.e., findings, results, or recommendations that could impact U.S. agriculture and food systems, if used;
 - b) intermediate outcomes, i.e., how the findings, results, or

recommendations have been used by others to influence practices, legislation, research/product design, training and so forth; and

- c) c. end outcomes, i.e., how findings, results, or recommendations have contributed to documented reductions in work-related morbidity, mortality, and/or exposure.

This project focused on basic research on an emerging system, so that its impacts are pretty much all potential outcomes. These were detailed in section 2.2; in brief, they are:

- Planting the right maize variety is more important than whether endophytes are applied
- Predicting the outcome of an endophyte treatment is complicated and depends on the maize variety, the endophyte, and the local environment.
- Plant tissue compartment appears to have the largest effect on microbes, followed by the local environment (soil, water, management, etc.), with genetics of the host plant playing a relatively minor role.
- Host plant genetics has very little effect across different environments, but can be a significant effect within a given environment.
- We found very little signal of heterosis in the maize microbiome.

- 2.4. What were the goals/specific aims of the project? If the approved application lists milestones/target dates for important activities or phases for this reporting period, identify these milestones and dates, as well as show actual completion dates or the percentage of completion of milestone targets. (up to 1,500-word limit)

Objective 1 (100% complete)

- Test beneficial endophytes on a select set of maize lines in the greenhouse
- Calculate broad-sense heritability of these interactions
- Determine relative genetic effects of host and endophyte
- Publish results

Objective 2 (95% complete; publication is currently under revision)

- Contact Genomes to Fields cooperators to identify 25-30 target sites for sampling
- Devise training materials for standardized collections
- Sample maize stalks and soil at all locations; extract & sequence
- Determine genetic and environmental drivers of stalk microbiome
- Publish results

Objective 3 (100% complete)

- Compare microbiomes of inbred and hybrid maize
- Quantify effects of tissue, environment, and heterosis
- Publish results



- 2.5. Have any of the major goals/specific aims or milestones for the project changed since the award? If so, please list the goal(s) that have changed and provide justification for the change from the approved goals. (up to 500-word limit)

Objective 1 was originally to screen a single beneficial endophyte against a large number of plants to perform genetic mapping. However, when starting to scale up to the actual objective, we found the effect smaller than our preliminary data indicated. We thus pivoted to look at multiple beneficial endophytes on a small number of plant varieties. This analysis was published in 2022.

Objective 3 originally aimed to perform selection on endophytic microbes to alter plant performance. Multiple attempts to carry this out failed around generation 3-4 due to plants becoming stunted, sick, and dying. We suspect that the experimental conditions were favoring pathogens, which reached a critical mass at this point in the experiment. Rather than continue to face this issue, the grad student on this project pivoted to make use of data we had gathered previously on the effect of maize heterosis on microbial communities. The samples for this had been processed but never fully analyzed, so the student took it to completion. It was published in April 2023.

- 2.6. What was accomplished under the goals/specific aims or milestones for the project? Please describe in detail, 1) major activities; 2) specific objectives; 3) significant results, including major findings, developments, or conclusions (both positive and negative); and 4) key outcomes or other achievements. Include discussion of stated goals not met. The emphasis in reporting in this section should focus on accomplishments. In the response, emphasize the significance of the findings to the scientific field. This section can be as technical as the PI would like. (up to 5,000-word limit)

1) Major Activities

- Objective 1 - Screen 3 different beneficial endophytes against a set of maize inbred lines to test for genome-by-genome interaction
- Objective 2 - Sample maize stalks from the Genomes to Fields initiative across the US and determine the genetic and environmental drivers of the stalk endophyte community.
- Objective 3 - Characterize the microbiome of inbred and hybrid maize to determine any heterotic effects on the maize microbiome.

2) Specific objectives

[These are the same as the major activities]

3) Significant results

These results are also reported in section 2.2 and are repeated for convenience

Objective 1 – Genetic variability of maize’s response to beneficial endophytes

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few organisms are able to make it into the maize stalk, and the community is relatively consistent across environments. This implies that (1) a beneficial microbe able to live in the stalk may do so consistently across environments, and (2) beneficial microbes that stay confined to the root or rhizosphere may be easier to develop since these compartments seem more amenable to a large number of diverse microbes.

4) Key outcomes and other achievements

- Training of 2 graduate students and 4 undergraduate students in bioinformatics, genetics, microbiology, and/or field analysis.
 - Both graduate students also secured summer internships with private-sector companies. One has since graduated and now works for Bayer; the other will be graduating in Dec 2023 and has lined up a postdoc in St. Jude's Children's Hospital.
- Presentation of these results at half a dozen different academic conferences
- Publication of 2 (soon 3) papers detailing the results reported above

2.7. Discuss efforts taken to ensure the approach is scientifically rigorous.
Include approaches taken to ensure robust and unbiased results.

Students are trained in scientific rigor and reproducibility. The PI consistently challenges students to verify their results through other means. Where experimental designs or analyses must make compromises between the idea and what is feasible, these are pointed out in publications. All informatic scripts and raw data are made available at time of publication.

2.8. List key stakeholders that could be served by results of this project.

Potential stakeholders:

- Crop producers, especially those interested in using beneficial microbes
- Biotech companies focusing on plant-microbe interactions
- Major plant breeding companies
- Other scientific researchers looking at plant-microbe interactions

2.9. How have the results of this reporting period been disseminated to communities of interest? Describe how the results have been disseminated to communities of interest. Include any outreach activities that have been undertaken to reach members of communities who are not usually aware of such activities, to enhance public understanding and increase interest in learning and careers in science, technology, and the humanities. Reporting the routine dissemination of information (e.g., websites, press releases) is not required. For awards not designed to disseminate information to the public or conduct similar outreach activities, a response is not required; the grantee should write "nothing to

report.” A detailed response is only required for awards or award components that are designed to disseminate information to the public or conduct similar outreach activities. Note that scientific publications and sharing research sources will be reported under Information Products. (up to 1,000-word limit)

Nothing to report

- 2.10. Describe challenges or delays encountered during project and actions that were taken to resolve them. Only describe significant challenges that may have impeded the research and emphasize their resolution. (up to 500-word limit)

Aside from the project pivots described in section 2.5, the largest obstacle during this time was the COVID-19 pandemic, as it presumably was for most researchers. It made all aspects of the project significantly more difficult and resulted in some considerable delays to some parts. We created work accommodations to permit work to continue during the pandemic, especially with allowing bioinformatic work to continue at home. Other accommodations included additional vehicles to help with fieldwork (so that fewer people were in a single vehicle at a time) and staggered shifts in the wet lab.

- 2.11. What opportunities for training and professional development has the project provided during this reporting period? If the research is not intended to provide training and professional development during this period, state “Nothing to Report.” For all projects reporting graduate student and/or post-doctoral participants, grantees are encouraged to describe how Individual Development Plans (IDPs) are used to help manage the training for those individuals. (up to 500-word limit)

This project trained 2 graduate students and 4 undergraduate students. All graduate students are required to create an Individual Development Plan and encouraged to revise it every 6 months. The PI goes over these plans at least annually to make sure the students are getting the preparation they want.

- 2.12. Please indicate the number of undergraduate and graduate students, post-doctoral scholars, or other educational components involved during this reporting period. If other education components are involved, please describe them in detail. (up to 500-word limit)

2 graduate students + 4 undergraduate students.



3. Information Products

- 3.1. Please list the type(s) of information products (e.g., scholarly publications, reports or monographs, workshop summaries or conference proceedings, video, audio, images, models, software, curricula, instruments or equipment, intervention, etc.) produced during the project resulting directly from the FFAR award.

Scientific publications:

- Li H, Griffis HK, Edwards J, Flint-Garcia S, Hirsh C, et al. **Relative contribution of genetics, environments, and GxE interactions on maize stalk endophytes.** (currently in revision)
- Schultz CR, Johnson M, Wallace JG. **Effects of Inbreeding on Microbial Community Diversity of Zea mays. Microorganisms.** (2023) Mar 29;11(4):879. doi: 10.3390/microorganisms11040879. PMID: 37110300; PMCID: PMC10145435.
- Corey R. Schultz; Kamaya M. Brantley; Jason G. Wallace (2022). **The role of genetic variation in Zea mays response to beneficial endophytes.** Plant Growth Regulation 98 (1): 167–177. DOI: 10.1007/s10725-022-00842-9.
- Giangacomo, Cecelia; Mohseni, Mohsen; Kovar, Lynsey L; Wallace, Jason. (2021) **Comparing DNA Extraction and 16S rRNA Gene Amplification Methods for Plant-Associated Bacterial Communities.** Phytobiomes Journal. DOI: 10.1094/PBIOMES-07-20-0055-R.

- 3.2. Please provide a list of citations for the information products produced during the project.

[See 3.1]

- 3.3. Are there publications or manuscripts accepted for publication in a journal or other publication (e.g., book, one-time publication and monograph) during the project resulting directly from the FFAR award? If yes, please provide citation.

[See 3.1]

- 3.4. Website(s). List the URL for any internet site(s) that disseminates the research activities. A short description of each site should be provided. It is not necessary to include the publications already specified above.

<https://wallacelab.uga.edu/> - Wallace Lab website

- 3.5. Have inventions, patent applications, and/or licenses resulted from the project (U.S. and international)? If yes, indicate the invention, patent application(s), and/or license(s).

	# Filed (Enter Numeric Value)	# Approved (Enter Numeric Value)	Patent Numbers (Separated by commas)
Patents	0	0	
Copyrights	0	0	
Trademarks	0	0	
Inventions	0	0	

- 3.6. Are any of the information products produced during this grant that are confidential, proprietary, or subject to special license agreements? If so, please list them below and describe why they must remain confidential. Also, note if (and when) you plan to make these data publicly available in the future or if they must remain confidential indefinitely. (up to 500-word limit)

No.

- 3.7. Beyond depositing information products in a repository, what other activities have you undertaken to ensure that others (e.g., researchers, decision makers, and the public) can easily discover and access the listed information products? What other activities have you undertaken to ensure that others can access and re-use these data in the future?

Methods are described in each publication, and the associated bioinformatic scripts are uploaded to a dedicated, public Github repository that is linked within each manuscript.

4. Data Management

- 4.1. Did the project generate any data? Data generation includes transformation of existing data sets and data from existing resources (e.g., maps and imagery). Please list the data generated for the award.

- High-throughput sequencing of maize microbial communities

- Field and greenhouse environmental data and corresponding maize performance
- 4.2. If you list multiple data sets, are these data sets related? If so, please provide a short description of how they are related.

The datasets are generally related according to the project they were from. For example, greenhouse environmental data and maize performance data were used to calculate the effect of different beneficial microbes across different maize varieties (Aim 1, published in Schultz et al 2022, above), with the same for Aims 2 and 3.

- 4.3. Please provide copies of relevant metadata records to support FFAR's mission of enhancing the discoverability of FFAR funded project data and information products. Include or attach copies of records and a simple file inventory, if necessary, in a compressed folder.

Datasets and metadata are published in standard repositories:

Publication	Datasets/Metadata	Github repository
Li H, Griffis HK, Edwards J, Flint-Garcia S, Hirsh C, et al. Relative contribution of genetics, environments, and GxE interactions on maize stalk endophytes. (in revision)	NCBI Proproject PRJNA883673	https://github.com/wallacelab/paper-li-g2f2019-microbiome
Schultz CR, Johnson M, Wallace JG. Effects of Inbreeding on Microbial Community Diversity of Zea mays. Microorganisms. (2023) Mar 29;11(4):879. doi: 10.3390/microorganisms11040879. PMID: 37110300; PMCID: PMC10145435.	NCBI Bioproject PRJNA924784	https://github.com/wallacelab/paper-schultz-microbiome-2023
Corey R. Schultz; Kamaya M. Brantley; Jason G. Wallace (2022). The role of genetic variation in Zea mays response to beneficial endophytes. Plant Growth Regulation 98 (1): 167–177. DOI: 10.1007/s10725-022-00842-9.	[supplemental files of publication]	https://github.com/wallacelab/paper-schultz-endophytes-2021
Giangacomo, Cecelia; Mohseni, Mohsen; Kovar, Lynsey L; Wallace, Jason. (2021) Comparing DNA Extraction and 16S rRNA Gene Amplification Methods for Plant-Associated Bacterial Communities. Phytobiomes	NCBI Bioproject PRJNA646931	https://github.com/wallacelab/paper-giangacomo-16s-methods



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Journal. DOI: 10.1094/PBIOMES-07-20-0055-R.		
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Certification

The undersigned hereby certifies that to the best of their knowledge and belief, the above report and all supporting documents are true, accurate and complete. I am aware there is a significant penalty for submitting false or misleading information.

If applicable, I agree that my electronic signature is legally binding, equivalent to, and has the same validity and meaning as my handwritten signature. I will not claim otherwise.

PI Full Name: Jason Wallace

PI Signature:  Date: 25 Oct 2023

Authorized Signing Official (ASO) Name: Hunter Hellwig

ASO Signature:  Date: 10/31/2023